Product (IP/IC) Code generation rule
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RS-COP-02-I-01 Revision history

Rev. no	Rev. date	Simple description of changed contents	Author	Post
1.0	2022.09.27	Newly established.	Mark.Kim	QM
1.1	2025.03.10	Product code update	Mark. Kim	QM
1.2	2025.10.30	IP Code generation rule update	Mark. Kim	QM
1.3	2025.11.11	4.6 MPW Top-cell name generation rule	Mark.Kim	QM

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1. Purpose

The purpose of this product (IP/IC) code generation rule is to establish a standardized naming and code assignment system for IP and IC developed or managed by the Ramschip. Inc. The code generation rule enables accurate identification, traceability, and classification across all development, sales, production, and quality assurance processes. This system also facilitates efficient communication and data management internally and externally with partners and customer.

2. Apply scope

This code generation rule applies to all IPs and products developed, acquired, or managed by the Ramschip, including analog, digital, PHY, and controller IPs, as well as integrated SoCs, test chips, and packaged products. It covers all documentation, DB, project folders, and system tools where codes are assigned and referenced.

3. Definition of terms

3.1 IP (Intellectual Property)

IP: A reusable design block of logic, circuit, or layout used in SoC development.

3.3 Product

Product: A physical or functional output derived from IP integration, including test chips, packaged ICs, evaluation boards, and customer deliverables components.

3.4 Code

Code: A structured alphanumeric identifier assigned to an IP or product for unique classification.

3.5 Version

A revision number or letter indicating design updates, bug fixes, or functional improvements within the same code family.

4. Product code generation rule

4.1 IP Code generation rule

(Category)_(Product symbol)(Product code)_(Process code)_(Configuration code) – All lower case

※ In the case of product symbol for sub-license IP, the defined name from Samsung Foundry should be used.

※ Configuration rules are not mandatory. It's just optional.

(Category)	(Product symbol)	(Product code)-4digit	(Process code)	(Configuration code)
(ramschip s-license IP) rs_ (ramschip own-IP) rc_	dcphy dphy cphy hsstptx hdptx etphy (ethernet phy) mtpphy aphy ipitx (display phy)	28nm: 1000 ~ 1999 14nm : 2000 ~ 2999 8nm : 3000 ~ 3999 5nm : 5000 ~ 5999 4nm : 6000 ~ 6999	ln28lpp ln14lpu ln08lpu ln05lpe ln04lpp - - -	m4s4 m0s22 n12e12 - - - -

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4.2 IP Code update rule

(Category)_(Product symbol)(Product code)(rev symbol)_(Process code) – All lower case

(Category)	(Product symbol)	(Product code)-4digit	(rev symbol)	(Process code)
(ramschip s-license IP) rs_ (ramschip own-IP) rc_	dcphy dphy cphy hsstptx hdptx etphy (ethernet phy) mtpphy aphy - - -	28nm: 1000 ~ 1999 14nm : 2000 ~ 2999 8nm : 3000 ~ 3999 5nm : 5000 ~ 5999 4nm : 6000 ~ 6999	a,b,c,d,e,f ------- ▶Except 'r'	ln28lpp ln14lpu ln08lpu ln05lpe ln04lpp - - -

4.3 Product code generation & revision rule

(Category)_(Product code #1)(Product code #2)(Version) – All upper case

Ex) RS5011A / RS5012A

(Category)	(Product code #1)	(Product code #2)	(Version)
(ramschip product) rs_	40: HSSTP 50: MIPI APHY Family ICs	11: RX 12: TX 13: RX/TX	A,B,C,D -----Z

4.4 Marking code generation rule (MPW project / Product based project)

(Category)	(Product code #1)	(Product code #2)	(Version)
(ramschip product) rs_	MPW Index Ex) MPW33	Process information Ex) LN08LPU	A,B,C,D -----Z (only product) In case of MPW project, N/A

→ In case of product based project, refer to the 4.3 rule.

4.5 Design Kit naming rule

(IP name)_(Classification#1)_ (Classification#2)_(YYMMDD)

※ Classification #2: Generate in uppercase / Configuration #1: Optionable

ex) rs_hsstpech_ln14lpp_v1p0_master_FE_230323

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(IP name)	Classification #1	Classification #2	YYMMDD
Defined IP name	master slave	FE BE DOC	230323

4.6 MPW Top-cell name generation rule

Ex) Q05N2601A

Description: Q(Quality verification) xxN(process node) YYMM(MPW Tape-out schedule) A(MPW GDS Index: A,B,C ---)

5. Relevant document

Standard	Document code
Development process	RS-COP-01

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